Sandeep Kumar Bollavaram

bollavaramsandeep@gmail.com | +91 63053 90913 | sandeepkumarbollavaram.in | github.com/sandeepbollavaram | linkedin.com/in/sandeepbollavaram

PROFILE

Full-stack, platform and AI engineer with **2,000+ GitHub contributions** and 20+ shipped projects spanning Kubernetes operators, LLM infrastructure, agentic developer tooling, offline-first AI and production web and mobile apps. Builds local-first AI systems with MCP, RAG and vector retrieval, and cloud-native platforms with Go, Kubebuilder and gRPC. Ships real deployments: staging.anythingdownload.in (Oracle Cloud) and ecellscsvmv.in. Currently freelancing at Algorithec on an AI-driven commerce decision engine with ONDC integration, leading a fintech complaint platform for Vastu Housing Finance, and heading a 16-member team as E-Cell SCSVMV President.

EDUCATION

SCSVMV University **2024 – 2028**

B.Tech in Computer Science and Engineering | ***CGPA: 7.88 / 10*** | ***Last Semester GPA: 8.08*** *Kanchipuram, Tamil Nadu*

EXPERIENCE

Algorithec Private Limited **Jun 2026 – Present**

Freelance AI/ML Engineer (joined as AI/ML Engineer Intern, Jun – Aug 2026) *Remote*

- Core contributor to Algorithec's AI-driven commerce decision engine (pre-launch), shaping its core architecture and working across both the backend services and the decision-engine layer.
- Led the engineering team on assigned initiatives during the internship phase, coordinating with teammates and the founder through standups, sprint planning and weekly reviews; continued as a freelancer through the product build.
- Owned the API integration with ONDC (Open Network for Digital Commerce) and built the firewall layer securing the commerce workflows.

PROJECTS

Complaint Intelligence Platform | Python, FastAPI, Next.js 14, SQLAlchemy, pytest **Vastu HFC, Prototype**

- Architect and tech lead of a pre-production platform for Vastu Housing Finance: omnichannel complaint intake including a 7-language voice agent, evidence-cited investigation, human-approved resolutions and a hash-chained audit log, with ~990 automated tests.

Anything Download | Next.js 16, FastAPI, Redis, FFmpeg, Docker, Oracle Cloud **Open Source, Live**

- Solo-built privacy-first media toolkit with **38 tools** across video, audio, image, PDF and web, plus a Chrome extension; a capability-driven engine inspects each link or file and offers only operations that genuinely apply.
- Next.js 16 frontend, FastAPI backend and a Redis job queue with separate worker and cleanup processes running FFmpeg and PDF pipelines; results are temporary and self-delete.
- Hardened against SSRF (private ranges, cloud metadata, redirects, DNS rebinding) with spoof-resistant rate limiting and strict upload, memory, timeout and output caps.
- Own domain, deployed on Oracle Cloud behind Caddy with automatic HTTPS at **staging.anythingdownload.in**; **377 automated tests** (73% backend coverage) and a 6-job CI pipeline with Trivy scans and dependency audits.

VORTEX | Go, AI Agents, Reverse Proxy/TLS, Docker, Telegram Bot API **Open Source**

- Self-hosted autonomous infrastructure platform in Go that provisions, monitors and self-heals services on a single node, with an integrated reverse proxy and automatic TLS certificate management.
- Multi-agent AI runtime that plans and executes infrastructure operations, controlled through a Telegram interface; 600+ commits across orchestration, health checks and deployment automation.

Kairo | MCP Protocol, TypeScript, Vector Memory, RAG, Agent Systems **Open Source**

- MCP-based engineering memory platform with 40+ tools for persistent context, repository intelligence, checkpoint recovery and agent continuity across AI coding workflows.
- Local-first vector retrieval and semantic memory that surfaces past decisions, architecture context and code knowledge without any cloud dependency.

ALOS | Python, PyQt6, Ollama, ESP32, WebSockets, SQLite, MCP, faster-whisper **Open Source**

- Fully offline personal AI assistant (no cloud, no API key) with a 3-panel desktop GUI and an 8-intent router covering chat, scheduling, reminders, health, code, smart home, planning and system tasks.
- Local voice I/O, a Python Skills plugin system, MCP client support (stdio and HTTP), and a WebSocket ESP32 bridge driving a 5-screen LVGL touchscreen and relays; 7-table SQLite layer and 160+ automated tests.

Shukra Operator | Go, Kubernetes, Kubebuilder, Helm OCI, gRPC, CRDs, Webhooks **Open Source**

- Production Kubernetes operator reconciling full-stack environments (Deployment, Service, Ingress, HPA, migration Jobs, backup CronJobs, NetworkPolicy) from a single AppEnvironment CRD.
- Mutating, validating and conversion webhooks, finalizer-driven deletion, Prometheus metrics, leader election and a gRPC bridge; 8 public releases shipped via OCI Helm charts.

Additional Projects: OCI-SentinelMesh (Python, OCI) compliance monitoring agent, in progress; University ERP (Next.js, Flutter, Firebase) deployed with QR attendance; Vayu (WinUI 3, .NET) offline Windows AI layer; Weapon Detection (YOLOv8, OpenCV, Flask); Vibrance (C#) photo editor; Pupixel Chrome extension.

TECHNICAL SKILLS

- **Languages:** Go, Python, TypeScript, C#, JavaScript, Dart, SQL, C++, R
- **AI / LLM:** LLM Applications, AI Agents, Agentic Workflows, RAG, Explainable RAG, MCP Protocol, Vector Databases (pgvector, ChromaDB), Embeddings, LangChain, Ollama, Local LLMs, Whisper, Human-in-the-loop AI
- **Platform:** Kubernetes, Helm, Kubebuilder, controller-runtime, CRDs, Webhooks, gRPC, Docker, GitHub Actions, Caddy, Trivy, CI/CD, Linux, Oracle Cloud (OCI)
- **Backend / Full Stack:** FastAPI, SQLAlchemy, Node.js, Express, REST, WebSockets, Redis Queues, Event-driven (Outbox), Next.js, React, Flutter, Tailwind CSS, pytest, Playwright
- **Data / Desktop / Hardware:** PostgreSQL, Redis, SQLite, Firestore, Supabase, .NET, WinUI 3, PyQt6, Raspberry Pi, ESP32, LiDAR, SLAM, IoT

LEADERSHIP & ACTIVITIES

Entrepreneurship Cell (E-Cell), SCSVMV University **2026 – Present**

President & Team Lead *Kanchipuram, Tamil Nadu*

- Lead and manage a **16-member core team** across event programming, outreach, sponsorship and campus-wide startup initiatives.
- Organised **4+ events** under E-Cell IIT Bombay's National Entrepreneurship Challenge (NEC), including YUKTI '26 (The Big Idea Pitch), SCSVMV's NEC campus round, where 15 startup teams pitched to an expert jury.
- Designed, built, deployed and maintain the official E-Cell SCSVMV website, live at **ecellscsvmv.in** (Next.js): events and results, blog, magazine, newsletter sign-up, contact and legal pages.

CERTIFICATIONS & ACHIEVEMENTS

- AWS AI Practitioner Challenge (course), Udacity (Apr 2026)
- **1st Place**, Apogee 2026 Technical Symposium (Neuromorphic Computing paper: LIF neurons, STDP, spiking neural networks), SCSVMV University
- **2,000+ GitHub contributions**; open-source maintainer across AI tooling, Kubernetes operators and developer tools; 20+ projects built
- **48 badges** earned on Microsoft Learn

SOFT SKILLS, INTERESTS & LANGUAGES

- **Soft Skills:** Leadership & Team Management, Communication & Public Speaking, Problem Solving, Event Planning & Coordination
- **Interests:** Tech writing on LinkedIn, Building side projects, Reading science & research
- **Languages:** Telugu (Native), English (Professional), Hindi (Conversational), Tamil (Basic)